Updated October 21st, 2026

James J. Griebler, Ph. D.

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APPOINTMENTS

Postdoctoral Research Associate, University of Tennessee, Knoxville, (August 2025 – present)
- *Future of Rheology* Invited Seminar Speaker (October 2025)

Graduate Research Assistant, University of Illinois Urbana-Champaign (August 2020 – July 2025)
- *Future of Rheology* Invited Seminar Speaker (April 2024)

Graduate Student Year-Round Intern, Sanda National Laboratory, (June 2023 – May 2025)

EDUCATION

Doctor of Philosophy, *Chemical Engineering* June 2025
University of Illinois Urbana-Champaign (UIUC), Urbana, Illinois

Master of Science, *Chemical Engineering* May 2024
University of Illinois Urbana-Champaign (UIUC), Urbana, Illinois

Bachelor of Science, *Bioproducts and Biosystems Engineering* May 2020
Bachelor of Science, *Mathematics*
University of Minnesota (UMN), Minneapolis, Minnesota

RESEARCH INTERESTS

- Recovery rheology
- Optical and scattering techniques for materials characterization
- Data-driven phenomenological model design
- Instrumentation and method development

PUBLICATIONS

Accepted

K. C. Bhaskar, **James J. Griebler**, S. A. Rogers, G. F. Christopher, *Understanding soft glassy rheology and yielding of particle laden interfaces through strain decomposition,* J. Col. and Int. Sci. (2026)

S. Kannapadi, …, **James J. Griebler**, S. A. Rogers, A. S. Kuenstler, *The Role of Catalyst Mobility and Concentration on the Linear Viscoelasticity of Associative Dynamic Covalent Networks*, Macromolecules (2026)

A. Khatiwada, …, **James J. Griebler**, A.M. Hubbard, R.D. Corder, *Effects of PLA/PHB Blend Ratio and Wood Flour Loading on the Melt Rheology and Thermomechanical Properties of Biobased Polymer Composites*, ACS Applied Polymer Materials (2026)

James J. Griebler, A.S. Dabo, E.M. Maczuga, and S.A. Rogers, *Resolving dual processes in complex oscillatory yielding,* Physical Review Letters, **135**, 058201 (2025)

K.M. Kamani, Y. Shim, **James J. Griebler**, …, and S.A. Rogers. *Linking structural and rheological memory in disordered soft materials*. Soft Matter, **21**, 750 (2025)

James J. Griebler, A.S. Tappan, S.A. Rogers, A.M. Grillet, and J.W. Kopatz, *Printability criterion and filler characteristics model for material extrusion additive manufacturing.* Additive Manufacturing, **99,** 104651 (2025)

PUBLICATIONS (CONT.)

James J. Griebler, A.S. Tappan, S.A. Rogers, J.W. Kopatz and A.M. Grillet, *Methodology to determine printability criteria of highly concentrated pastes through rheological characterization.* Additive Manufacturing, **96,** 104578 (2024)

Y. Shim, **James J. Griebler**, S.A. Rogers, *A reexamination of the Cox-Merz rule through the lens of recovery rheology.* Journal of Rheology, **68**, 381–396 (2024)

R.N. Gaines, B.A. Kleimenhagen, **James. J. Griebler**,… , P.J.A. Kenis, *Optimizing the Flow Electrooxidation of Glycerol Using Statistical Design of Experiments*. Journal of The Electrochemical Society, 171, 063506 (2024)

James J. Griebler, G.J. Donley, V. Wisniewski, S.A. Rogers, *Strain shift measured from stress-controlled oscillatory shear: evidence for a continuous yielding transition, and new techniques to determine average recovery measures.* Journal of Rheology, **68**, 301–315 (2024)

J. M. While, A. Garza, **James J. Griebler,** F.S. Bates, M. A. Calabrese, *Engineering the Structure and Rheological Properties of P407 Hydrogels via Reverse Poloxamer Addition*, Langmuir, **39**, (2023)

James J. Griebler, S.A. Rogers. *The nonlinear rheology of complex yield stress fluids.* Physics of Fluids, **34,** 0203107 (2022)

In Preparation / Submitted

James J. Griebler, … B. G. Compton, R. D. Corder, *Reconciling FTIR, DSC, and Rheology Measurements of Cure in Ceramic-Filled Preceramic Polymer Systems,* (Under Review)

James J. Griebler, N. Smith, G. Rother, S. Mohammadkhah, S. Khan, R. D. Corder, *Structural and Rheological Effects of Temperature and Carbohydrate Concentration on Denatured Whey Protein Isolate Hydrogels,* (In Revision)

C. H. Lindeman*, James J. Griebler*, …, R. H. Leheny, Heterogeneous Relaxational Dynamics and Evolution of Recoverable Strain following Cessation of a Nanocolloidal Soft Glass, (Under Review)

James J. Griebler, S. A. Rogers, *Recovery Rheology: A Perspective*, (In Preparation)

PROPOSALS AND FUNDING

Funded/Awarded

- UT Knoxville Center for Materials Processing SEED Grant
- Oak Ridge National Lab, Spallation Neutron Source Beamtime Proposal IPTS-37896.1
- Oak Ridge National Lab, Spallation Neutron Source Beamtime Proposal IPTS-35473.1
- Argonne National Lab, Advanced Photon Source Beamtime Proposal Experiment #277635

PRESENTATIONS, PROCEEDINGS & POSTERS

* Indicates speaker

James J. Griebler*, N. Smith, G. Rother, S. Mohammadkhah, S.A. Khan, K. M. Weigandt, R. D. Corder, *Linking Network Architecture to Mechanical Response in Protein–Polysaccharide Hydrogels via Neutron Scattering and Rheology*, 100th Annual ACS Colloids Meeting, June 23rd, 2025, Newark, DE (Oral Presentation)

PRESENTATIONS, PROCEEDINGS & POSTERS (CONT.)

James J. Griebler, C. Mortensen, B. G. Compton, R. D. Corder*, *In-Situ Characterization of Photocure Kinetics in Ceramic-Filled Pre-Ceramic Polymers via Rheology, FTIR, and DSC*, 100th Annual ACS Colloids Meeting, June 24rd, 2025, Newark, DE (Oral Presentation)

Chloe W Lindeman*, Sidney R. Nagel, **James J. Griebler,** Penelope G. Kovakas, Simon A. Rogers, Suresh Narayanan, James L. Harden, Robert L. Lehaney, *Memory in amorphous solids: From micro- to macro-rheology*. The Society of Rheology 96th Annual Meeting, October 20th, 2025, Santa Fe, New Mexico (Oral Presentation)

James J. Griebler*, G.J. Donley, V. Wisniewski, S.A. Rogers. *Stain shift measured from stress-controlled oscillatory shear: evidence for a continuous yielding transition, and new techniques to determine average recovery measures.* **Invited talk**, The Society of Rheology 96th Annual Meeting, October 21th, 2025, Santa Fe, New Mexico (Oral Presentation)

James J. Griebler, Jessica W. Kopatz, S.A. Rogers, Alex S. Tappan, Anne M. Grillet*, *Printability Criterion for highly filled inks for direct-ink write additive manufacturing.* The Society of Rheology 96th Annual Meeting, October 21th, 2025, Santa Fe, New Mexico (Oral Presentation)

James J. Griebler, A.S. Tappan, S.A. Rogers, A.M. Grillet, J.W. Kopatz*, *Printability Criterion of Highly Filled Pastes for Direct-Ink Writing Based on Small Amplitude-Oscillatory Shear (SAOS).* 2024 AIChE Meeting, October 28th, 2024, San Diego, CA. (Oral Presentation)

James J. Griebler*, …, S.A. Rogers, *An explanation of double overshoot yielding.* The Society of Rheology 95th Annual Meeting, October 17th, 2024, Austin, TX. (Oral Presentation)

James J. Griebler*, G.J. Donley, V. Wisniewski, S.A. Rogers, *Stain shift measured from stress-controlled oscillatory shear: evidence for a continuous yielding transition.* The Society of Rheology 95th Annual Meeting, October 16th, 2024, Austin, TX. (Poster Presentation)

K.M. Kamani*, Y. Shim, **James J. Griebler**, S. Narayanan, Q. Zhang, R.L. Leheny, J. Harden and S.A. Rogers*. *Linking structural and rheological memory in disordered soft materials*. The Society of Rheology 95th Annual Meeting, October 16th, 2024, Austin, TX. (Oral Presentation)

James J. Griebler*, A.S. Tappan, S.A. Rogers, A.M. Grillet, J.W. Kopatz, *Printability Criterion of Highly Filled Pastes for Direct-Ink Writing Based on Small Amplitude-Oscillatory Shear (SAOS).* The Society of Rheology 95th Annual Meeting, October 16th, 2024, Austin, TX. (Oral Presentation)

James J. Griebler*, G.J. Donley, V. Wisniewski, S.A. Rogers. *Stain shift measured from stress-controlled oscillatory shear: evidence for a continuous yielding transition, and new techniques to determine average recovery measures.* **Invited talk**, Society of Rheology Future of Rheology Seminar Series, April 25th, 2024, Virtual. (Oral Presentation)

James J. Griebler*, J. Kopatz, S.A. Rogers, A.S. Tappan, A.M. Grillet. *Methodology to determine filler-agnostic printability criteria for DIW inks through small amplitude oscillatory shear*. 2024 Colloidal, Macromolecular and Polyelectrolyte Solutions Gordon Research Conference, February 16th, 2024, Ventura, CA. (Poster Presentation)

James J. Griebler*, J. Kopatz, S.A. Rogers, A.S. Tappan, A.M. Grillet. *Methodology to determine filler-agnostic printability criteria for DIW inks through small amplitude oscillatory shear.* 2024 Colloidal, Macromolecular and Polyelectrolyte Solutions Gordon Research Symposium, February 10th, 2024, Ventura, CA. (Poster Presentation)

K.M. Kamani, Y. Shim, **James J. Griebler**, S. Narayanan, Q. Zhang, R.L. Leheny, J. Harden and S.A. Rogers*. *A Generalized Memory Function Based on Recoverable Strain and Its Relation to Nanoscale Structure.* 2023 AIChE Annual Meeting, November 6th, 2023, Orlando, FL. (Oral Presentation)

J. White, A. Garza, **James J. Griebler**, F. S. Bates, M.A. Calabrese*. *Engineering the Structure and Rheological Properties of P407 Hydrogels Via Reverse Poloxamer Addition.* 2023 AIChE Annual Meeting, November 6th, 2023, Orlando, FL. (Oral Presentation)

K.M. Kamani*, Y. Shim, **James J. Griebler**, …, S.A. Rogers. *A Generalized Memory Function Based on Recoverable Strain and Its Relation to Nanoscale Structure.* 9th Viscoplastic Fluids Workshop, October 11th, 2023, Monticello, IL (Oral Presentation)

James J. Griebler*, ..., S.A. Rogers, *Stain shift measured from stress-controlled oscillatory shear: evidence for a continuous yielding transition.* 9th Viscoplastic Fluids Workshop, October 9th, 2023, Monticello, IL. (Oral Presentation)

J. Kopatz*, **James J. Griebler,** A. Cazares**,** J. Leonard, A.S. Tappan & A.M. Grillet. *Accurate Rheological Characterization of Highly-Filled Direct-Ink Write Pastes*. XIXth International Congress of Rheology, July 31st, 2023, Athens, Greece. (Oral Presentation)

James J. Griebler, …, S.A. Rogers*. *What recovery teaches us about yielding and double yielding.* XIXth International Congress of Rheology, July 31st, 2023, Athens, Greece. (Oral Presentation)

James J. Griebler*, G. J. Donley, V. Wisniewski, S.A. Rogers. *Strain shift in a model yield stress fluid – evidence for a continuous yielding transition.* The Society of Rheology 93rd Annual Meeting, October 10th, 2022, Chicago, IL. (Oral Presentation)

TEACHING EXPERIENCE

Guest Lecturer, Rheology Graduate Course, University of Tennessee, Knoxville (March 2026)

Graduate Teaching Assistant, Chemical and Biomolecular Engineering, University of Illinois Urbana-Champaign (January 2022 – May 2023)

- Head TA in charge of running weekly discussions and grading for Mass and Energy Balances Course
- Lead TA in charge of Senior Unit Operations Lab course, helping students with experiment development and data analysis

Undergraduate Teaching Assistant, Mathematics and Bioproducts Engineering Departments, University of Minnesota Twin Cities (January 2018 – May 2020)

- Grader for College Algebra
- TA for Mass and Energy Balances

MENTORING EXPERIENCE

Graduate Students – 3 total (August 2022 – Present)

- Mentored 2 Ph. D. and 1 Masters student in the Rogers group, supporting research development, milestone exam preparation, and fellowship applications
- Mentored 2 Ph D. students in the Corder group, supporting research, fellowship, and grant applications
- Outcomes: students advanced through major program milestones (preliminary & qualifying exams) and earned competitive recognitions.

Undergraduate Students – 3 total (August 2022 – present)

- Mentored 3 students in rheological characterization projects
- Taught 3 students how to develop scientific posters and presentations
- Guided 2 students through graduate applications
- Outcomes: 2 student have been accepted to graduate school

SERVICE

Session Co-Chair, 2026 AIChE Annual Meeting, *Minneapolis, Minnesota*

- Facilitated abstract submission and acceptance for the Nonlinear and Turbulent flows session at the AIChE Annual Meeting
- Chairing individual sessions at the meeting

Session Co-Chair, The Society of Rheology 96th Annual Meeting, *Santa Fe, New Mexico*